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Structural Similarity as Guidance in Case-Based Design *

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Abstract. This paper presents a novel approach to determine structural similarity as guidance for adaptation in case-based reasoning (CBR). We advance structural similarity assessment which provides not only a single numeric value but the most specific structure two cases have in common, inclusive of the modification rules needed to obtain this structure from the two cases. Our approach treats retrieval, matching and adaptation as a group of dependent processes. This guarantees the retrieval and matching of not only similar but adaptable cases. Both together enlarge the overall problem solving performance of CBR and the explainability of case selection and adaptation considerably. Although our approach is more theoretical in nature and not restricted to a specific domain, we will give an example taken from the domain of industrial building design. Additionally, we will sketch two prototypical implementations of this approach.

1 Introduction

The effectiveness of case-based reasoning depends on the ability to determine former experiences (cases) that are useful and applicable to solve new, similar problems. When one tries to handle *synthesis* tasks as opposed to *analysis* tasks, however, the determination of similarity alone is not enough: The *adaptability* of former cases to problems of current interest becomes essential.

Most approaches to similarity assessment in CBR (c.f. [Kol92, Ric92]) estimate the usefulness of cases based on the presence or absence of certain features. The features are preclassified as important with respect to retrieval. Similarity is assessed by a numeric computation and results in a single number which is meant to reflect all aspects of the similarity. There are approaches (e.g., [Hin92]) which try to capture the plausible inferences intrinsically through the organization or *indexing* of knowledge. Constraints on a problem serve as indices into a design memory. The memory returns cases that provide a solution, some of the

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以结构相似性为指导在基于案例的设计中 *

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摘要。本文将这种paper presents a novel approach to determine structural相似性作为案例推理（CBR）中适应过程的指导依据。我们提出了一种结构相似性评估方法，该方法不仅能够给出一个single数值，还能确定两个案例之间具有in共性的最具体结构，从而揭示inclusive of the modification rules needed to obtain this structure from这两个案例之间的关联。我们的方法将检索、匹配与适应视为一组相互依赖的过程，这确保了在retrieval和匹配过程中能够识别出not only similar但同时具备适应性的案例。这两者相结合，既提升了案例推理的整体问题求解性能，又增强了其可解释性of case selection以及适应过程considerably.的透明度。尽管我们的方法在本质上更具理论性且不局限于某一特定specific领域，但我们will仍将提供一个源自工业建筑设计领域的实例。此外，我们还将简要介绍两种this approach.的原型实现。

1 引言

基于案例的推理的有效性取决于能否识别出那些对解决新的、类似问题具有参考价值且适用的过往经验（案例）。然而，当处理综合任务而非分析任务时，仅凭相似性判断是不够的：过往案例对当前关注问题的适应性变得至关重要。

案例推理中用于相似性评估的大多数方法（参见[Kol92, Ric92]）均基于某些特征的有无来估计案例的有用性。这些特征在检索过程中被预先划分为重要特征。相似性通过数值计算进行评估，并最终得到一个单一的数值，用以综合反映相似性的各个方面。此外，也有一些方法（例如[Hin92]）试图通过知识的组织或索引来内在地捕捉合理的推理过程。问题中的约束条件被用作设计记忆的索引，该记忆会返回能够提供解决方案的案例，其中部分

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context of the design as well as feedback or external evaluation. This information is used to determine how applicable the case is, how to adapt it, and how to avoid repeating previous failures. Contrary to this, we want to introduce an approach to similarity assessment which depends not only on prior cases but also on the domain background knowledge available for adaptation. Structural similarity is defined as the most common structure of cases and the rules needed to determine it. It is able to guide solution adaptation.

Conventional CBR systems which use causal models (e.g., CASEY [Kot88], SWALE [KLO86], CHEF [Ham86]) treat retrieval, matching (or justification), and adaptation separately one after each other. By contrast, we will show that the integration of these stages improves the suitability and performance of CBR significantly.

The paper is organized as follows: First, we introduce a novel definition and use of structural similarity in CBR and its motivation by prior work. Second, to exemplify our approach, we solve a specific synthesis task taken from the domain of building design. Third, we sketch two implementations of this methodology. We conclude by discussing our approach and delineate future work.

2 Our Approach: Structural Similarity as Guidance

To introduce our general approach to similarity assessment and adaptation, we use Fig. 1. As a basis to assess similarity in terms of adaptability a canonical system of cases and associated case modification rules[2] is needed. Prior cases have to be represented both in attribute-based and structural forms (e.g., by terms or graphs) and are stored in the case-base on the right side of Fig. 1. Background knowledge is represented by domain-dependent and task-dependent rules (e.g., term or graph modifications like generalizations, geometrical transformations, and substitutions) including their 'inverse' rules. This knowledge is specific for classes of cases and will be stored separately in the rule-base shown in the middle of the figure. Changes in case or background knowledge influence both similarity assessment and adaptation.

Structural similarity assessment is a computationally expensive process. Inspired by the work of DEDRE GENTNER and KENNETH D. FORBUS [GF91], we will tune computationally cheap, fast preselection and expensive structural similarity assessment to complete each other. The procedure (marked by arrows) is as follows.

Given the new problem in attribute-based description, we start by determining a set of candidate cases, the surface attributes of which are similar to those of the new problem. Based on this computationally cheap analysis (*surface similarity assessment*) of the problem, we can now use the transformation function ϕ to translate the new problem into a structural representation. Corresponding to the new problem and the preselected candidate cases, modification rules f_1, f_2 will be chosen (*rule selection*) and applied until a common structure of the actual

[2] The rules used here are not covered by the expression rewriting rules.

在设计过程中，还需结合反馈或外部评估。这些信息用于判断案例的适用性、适应方式以及如何避免重蹈覆辙。与此不同，我们拟提出一种相似性评估方法，该方法不仅依赖于已有案例，还充分考虑可用于适应的领域背景知识。结构相似性被定义为案例中最常见的结构，以及用于确定可指导解决方案适配的规则。

传统的基于案例的推理系统采用因果模型（如CASEY [Kot88]、SwWALE [KL.O86]、CHEF [Ham86]），将检索、匹配（或论证）与适应这三个阶段依次独立地进行。相比之下，我们将证明，将这些阶段加以整合能够显著提升基于案例的推理系统的适用性和性能。

本文的结构安排如下：首先，我们提出了在案例推理中关于结构相似性的新定义及其使用方法，并结合已有研究阐述其理论依据；其次，为说明所提方法的有效性，我们以建筑设计领域中的一个specific综合任务为例进行求解；再次，我们概述了该方法论的两种具体实现方案。最后，我们对所提方法进行了讨论，并展望了未来的研究方向。

2 我们的方法：以结构相似性为指导

为介绍我们用于相似性评估与适应的一般方法，我们以图1为例。为了从可适应性的角度评估相似性，需要一个规范化的案例体系及其相应的案例修改规则。先前的案例须同时以基于属性的形式和结构化形式（例如，通过术语或图）加以表示，并存储于图1右侧的案例库中。背景知识则由领域相关及任务相关的规则（例如，术语或图的修改操作，如泛化、几何变换和替换）及其“逆向”规则来表征。这些规则具有案例类别的特异性，将单独存储于图中所示的规则库中。案例或背景知识的变化均会同时影响相似性评估与适应过程。

结构相似性评估是一项计算开销较高的过程。受DEDRE GENTNER和KENNETH D. FORBUS [GF91]的研究工作启发，我们将对计算成本较低、速度较快的预筛选与计算成本较高的结构相似性评估进行协同优化，使其相互补充。具体流程（以箭头标识）如下。

鉴于基于属性描述的新问题，我们首先确定一组候选案例，其表面属性与新问题的表面属性相似。基于这一计算成本较低的问题分析（表面相似性评估），我们现在可以利用变换函数¢将新问题转化为结构化表示。针对新问题及预先选定的候选案例，将选择并应用相应的修改规则·、·，直至获得实际问题的共同结构a。

此处所使用的规则不属于表达式重写规则的范畴。

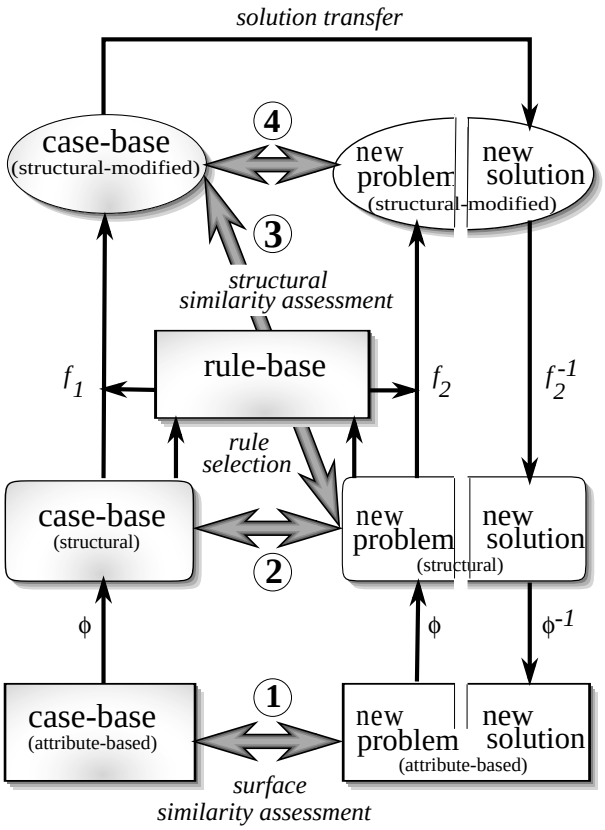


Fig. 1. General Approach and Different Levels to Assess Similarity

problem and one candidate case is found (*structural similarity assessment*). Note that $f_1 \neq f_2$ is possible. Now the solution of this candidate case, also modified, can be transferred to the new problem (*solution transfer*). After that, inverse modification rules f_2^{-1} are applied to get the concrete structural representation of the new solution. Using ϕ^{-1} we will get the attribute-based representation of the new solution[3]. This will be offered pictorially to the user.

The common structure of cases together with the modification rules applied to obtain them determine which prior solutions are useful. The inverse modification rules will show how to adapt them.

We should mention that similarity assessed by the system can considerably differ from that assessed by human users, even assuming that both have the

[3] In mathematical terms, it is not always possible to invert the mapping ϕ.Thus ϕ^{-1} denotes any computable flattening that takes a structural case description and transforms it into an attribute-based one such that applying ϕ would yield the structured representation, again. The same applies to f and f^{-1}.

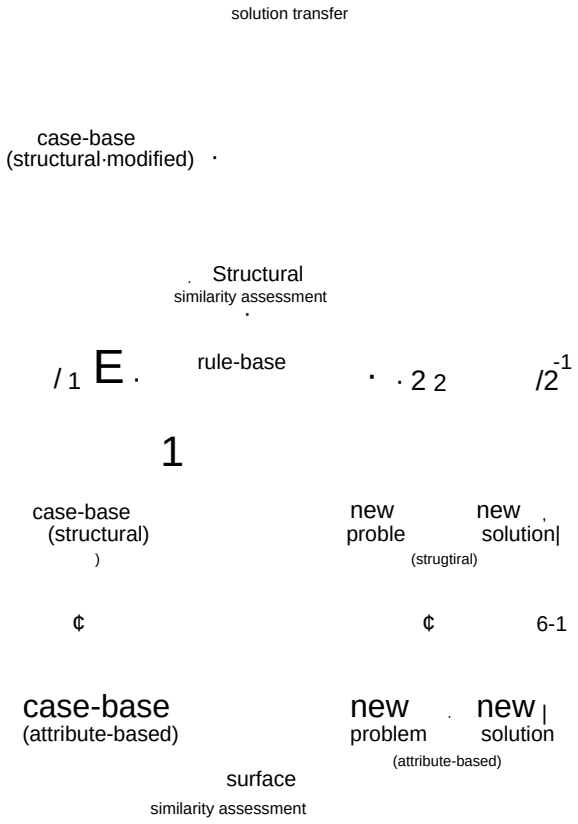


图1. 评估相似性的总体方法及不同层次

问题被识别，并找到了一个候选案例（结构相似性评估）。需要注意的是，此时，经过修改的候选案例$_{this}$的解可以被迁移至新问题中（解的迁移）。随后，应用逆向修改规则·1，以获得具体的结构表示of the new solution. Using b-1 we will get the attribute-based representation of the new solution®. This will be offered pictorially to the user.。

在应用修改规则以获得$_{cases\ together}$时，其通用结构决定了哪些已有解是可用的。相应的逆向修改规则将阐明如何对这些解进行适配。

应当指出，即使假设两者都具有……，由该系统评估的相似度也可能与人类用户评估的相似度存在显著差异。 e

¢从数学角度来看，$_{it}$ 并非总能对映射进行逆变换¢。因此，denotes any computable flattening that takes a $_{structural}$ case description and transforms $_{it\ into}$ an attribute-based one such that applying ¢ would yield the structured representation, again. The same applies to f and ·。

same set of cases at hand. Due to the background knowledge available their strategies to use and reason from these cases can be quite different. In each case the adapted solution should meet the given requirements.

Before giving a practical example we want to contrast our approach to prior work. In addition Fig. 1 presents conventional approaches to similarity assessment in CBR and analogy (denoted by fat grey double arrows and enumerated (1) to (4)):

(1) Frequently used in CBR are *surface similarity assessments* based on attribute-based case representations (c.f. [Weß91, Ric92, Kol92]). They provide a single numeric value which is meant to reflect all aspects of similarity. If similarity is defined depending on some given distance it inherits the properties of reflexivity, symmetry, and triangular inequality. This is sometimes not desirable. An excellent discussion about this topic provides [Jan93].

(2) If interdependencies of attributes have to be taken into account, representations like terms and graphs are used as a basis for similarity assessments. Again, the output is a single numeric value.

(3) There are approaches where cases stored in the case-base will be modified (e.g., using letter substitution rules as known in speech recognition) to determine similarity. Similarity is defined by the transformations used to transform one case into another.

(4) The principle of redescription [Ind91, O'H92] modifies new and old problem descriptions in a mutually dependent way in order to synthesize an isomorphism of both descriptions.

We emphasize that all the approaches mentioned above define similarity independent on the adaptation knowledge available.

3 An Example: Case-Based Industrial Building Design

Much work has been done in case-based building design (c.f. [Goe89, DK92, Nav92, HF93]), which is one of the most complex real world synthesis tasks. Architects experience the world in cases. CBR seems to be the natural problem solving method. Synthesis tasks in building design usually aim at the creation of objects and their conflict free arrangement corresponding to certain requirements.

In our project, we focus on the installation of supply system nets in industrial buildings with a complex infrastructure. Fig. 2 illustrates the layout of subsystems for return air. Return air accesses have to be connected by return air connections. The ellipses used in the figure provide information on a sketch level of design. To use ellipses instead of rectangles is a very useful graphical trick: Ellipses overlap only in a few points. Thus, information on different levels of abstraction can be displayed simultaneously. For a detailed introduction into this representation scheme, the reader is referred to [Hov93]. In Fig. 2, thinly drawn circles denote places where accesses can be placed. Bold drawn circles

同一组案例同时呈现。由于所具备的背景知识不同，他们利用这些案例并据此进行推理的策略可能大相径庭。在每一种情况下，所采用的适应性解决方案均应满足既定需求。

在给出一个实际示例之前，我们希望将我们的方法与先前的研究进行对比。此外，图1展示了案例推理中用于相似性评估的常规方法以及类比推理的方法（以粗灰色双箭头表示，并进行了编号）。

(1)至(4))：

(1) 在基于案例的推理中，经常采用基于属性的案例表示来进行表面相似性评估（参见[Wef91, Ric92, Kol92]）。此类方法给出一个单一的数值，用于反映相似性的某些方面。若相似性是依据某种给定的距离函数来定义的，则该距离函数将继承自反性、对称性和三角不等式等性质。然而，有时这种性质并不 desirable。关于这一问题的深入讨论可见[Jan93]。 (2) 当需要考虑属性之间的相互依赖关系时，通常以项和图等表示形式作为相似性评估的基础。同样，其输出结果也为一个数值。 (3) 还有一些方法通过对案例库中的案例进行修改（例如，采用语音识别中常见的字母替换规则）来确定相似性。此时，相似性由将一个案例转换为另一个案例所使用的变换过程来定义。 (4) 重描述原理[Ind91, O·H92]通过以相互依赖的方式同时修改新旧问题描述，从而实现两种描述之间的同构映射。

我们强调，上述所有方法均在不依赖于现有适应性知识的情况下定义相似性。

3 一个示例：基于案例的工业建筑设计

在基于案例的建筑设计领域已开展了大量研究工作（参见 [Goe89, DK92, Nav92, HF93]），该领域是现实世界中最复杂的综合任务之一。建筑师通过案例来感知和理解世界，因此案例推理似乎是一种天然的问题求解方法。建筑设计中的综合任务通常旨在根据特定需求创建各类建筑要素，并对其进行无冲突的布局规划。

在本项目中，我们重点关注具有复杂基础设施的工业建筑中送风系统管网的安装。图2展示了回风子系统的布置情况。回风口需通过回风连接管进行连接。图中所采用的椭圆符号在设计草图层面提供了相关信息。使用椭圆而非矩形是一种非常实用的图形化技巧：椭圆之间仅在少数点上发生重叠，因此可在同一图中同时呈现不同抽象层次的信息。有关该表达方案的详细说明，请参阅文献[Hov93]。在图2中，细线绘制的圆圈表示可设置风口的位置，而粗线绘制的圆圈

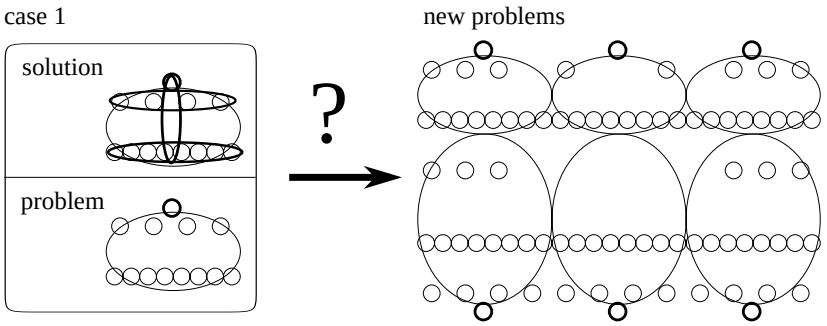


Fig. 2. Case-Based Design of Air Supply Nets

denote places reserved for main accesses. Ellipses denote areas where connections of supply accesses can be placed. The task to be tackled is the design of the connections for supplies that properly cover all of a given set of accesses for supply. Simply speaking, circles have to be covered by ellipses. The question is, how to transfer the experience stored in case 1 to the new problems at hand ? Identical problems occur seldom in design. Adaptation is essential. Using this task of connecting supply accesses, we will show how our approach works.

3.1 An Example: Knowledge Representation

The selection of an appropriate knowledge representation is strongly task and domain dependent. To realize our approach we need two different types of case representations as well as a complementing rule-based representation of proper case modifications.

The *attribute-based* case representation provides visually prominent features of objects. Following the work of LUDGER HOVESTADT [Hov93] each object (circle or ellipse) will be represented by its spatial dimensions and nine further attributes like *time* at which this object was created, *aspect* which assumes one of 'return air,' 'fresh air,' etc., and *morphology* which refers to 'access,' 'connection', etc. This fixed set of dimensions will be used as indices[4]. This representation permits us to use computer drawings as the main basis for man-machine interaction in building design. Architects can mark problems by simply manipulating objects in drawings. Solutions will be offered as drawings, too. Filling in large forms to represent each case or to describe a new problem will not work in real world applications.

Next, we have to encode case-based *structural* knowledge about relative positions and spatial arrangements of objects in a machine–usable form. Influenced

[4] To get cases in a less redundant form, 'space-coordinates' are normalized. Therefore the smallest x-, y-, and z-dimensions of each case are set to zero.

案例1 new problems

解决方案

问题



图2. 基于案例的送风管网设计

用圆圈表示主接入点的预留位置，用椭圆表示可布置供应接入点连接的位置。待解决的任务是设计供应接入点的连接方案，以确保其能够合理覆盖给定的一组供应接入点。简而言之，就是用椭圆来覆盖圆圈。问题在于……

是,

如何将案例1中积累的经验迁移到当前面临的新问题上？在设计过程中，完全相同的问题很少出现，因此适应性至关重要。通过这一连接供电接入点的任务，我们将展示我们方法的运作机制。

3.1 一个示例：知识表示

恰当的知识表示形式的选择在很大程度上取决于具体任务与领域。为实现我们的方法，我们需要两种不同类型的案例表示，以及一种用于描述适当案例修改的补充性规则基表示。

基于属性的案例表示能够直观地呈现对象的显著特征。参照LUDGER HOOVESTADT的研究成果[Hov93]，每个对象（圆形或椭圆形）将由其空间尺寸以及另外九个属性来表征，例如该对象的创建时间、表示“回风”“新风”等状态的气流属性，以及表示“入口”“连接”等形态属性。这一固定的维度集合将被用作索引4。这种表示方法使我们能够在建筑设计的人机交互中以计算机绘图为主要依据。建筑师只需通过操作图纸中的对象即可标记问题，而解决方案也将以图纸的形式给出。在实际应用中，采用填充大块区域的方式来表示每个案例或描述新问题的做法并不可行。

接下来，我们必须以机器可读的形式对关于物体相对位置与空间布局的基于案例的结构化知识进行编码。受……的影响

4为了使各案例形式更加简洁，对空间坐标进行了归一化处理。因此，每个案例的最小x、y和z维度均被设为零。

by the work of BIPIN INDURKHYA (c.f. [Ind91]), we represent the complex structures like supply air net structures as terms over some appropriately tailored signature. Therefore, a finite, heterogeneous, and finitary signature is assumed. This is taken as a basis for building terms and formulae, as usual. The detailed formal description of the signature used to represent cases structurally can be found in [Bör93]. Equational knowledge about functions and their relations is formalized . Note that each solution description contains the corresponding problem description. There is a function ϕ with its inverse which realizes the transformation of the attribute-based descriptions into structural ones and vice versa.

Additionally, we need knowledge (*rules*) about proper modifications of structural case representations. Terms can be modified using generalizations. To express generalized terms we need a sorted family of variables. For simplicity, we assume all variables to be called x, with indices whenever necessary. There are meaningful geometrical transformations like *reflection, rotation, translation*, etc. in our domain[5]. Additionally, structural representations can be modified using abstraction rules, which assign term expressions to constants (abstract attributes) like *row, regular, covered* etc. These three different kinds of modification rules including their inverses will be stored in the rule-base.

This knowledge provides a canonical system and the potential for structural similarity assessment which guides adaptation.

3.2 An Example: Similarity Assessment

For illustration, the main procedure given in Fig. 1 is exemplified in Fig. 3. The left, lowest box shows the pictorial and attribute-based representation[6] of one typical case stored in the case-base. By taking the functions *cover* and *copy* and object constants, combined with appropriate parentheses and commas, we are able to express the solution of this case structurally by *cover(copy(Y,3,Circle))*. This term stands for *take one circle, copy it three times and arrange them all in the y-direction. Afterwards cover all circles with a single ellipse.* The right, lowest box shows the pictorial and attribute-based representation of the new problem to be solved. Given in the same box but not available at this time is the solution of the new problem. The particular intention (also called subgoal) the user wants to concentrate on is the connection of air supplies.

The first initial analysis of the new problem can be done by an inexpensive *surface similarity assessment*, based on the attribute-based descriptions. The result is a set of candidate cases which have similar surface attributes such as aspect, number of objects etc. In this way, the rather large set of cases stored in the case-base can be reduced to a few useful candidate cases.

[5] In contrast to, e.g., mechanical engineering where functional dependencies constitute the background knowledge model, in building design topological dependencies play the paramount role and have to be considered during reasoning. Additionally, geometrical laws (reflecting a reflected figure again will result in the original figure) can be exploited.

[6] For simplicity, we only gave the values of the attributes x, dx, y, dy, time, aspect, and morphology of each object.

根据BIPIN INDURKHYA的工作（参见[Ind91]），我们将诸如送风管网结构之类的复杂结构表示为基于某一适当定制的符号集的项。因此，我们假定存在一个有限的、异构的且为有限型的符号集1s，并将其作为构建项与公式的基础，一如往常。用于结构性地表示各类案例的符号集的详细形式化描述可见于[Bor93]。关于函数及其相互关系的等式知识被形式化地表达。需要注意的是，每份解决方案的描述中均包含相应的问题描述。存在一个函数¢及其逆函数，它们实现了从基于属性的描述到结构性描述的转换，反之亦然。

此外，我们还需要关于结构化案例表示适当修改的知识（规则）。术语可以通过泛化进行修改。为了表达泛化的术语，我们需要一个带排序的变量族。为简便起见，我们约定将all variables to记为2，并在必要时使用下标。在我们的领域中，存在诸如反射、旋转、平移等有意义的几何变换®。此外，结构化表示还可以通过抽象规则进行修改，这些规则将项表达式赋予常量（抽象属性），如行、正则、覆盖等。这三类不同的修改规则及其逆规则都将存储在规则库中。

这些知识提供了一个规范化的系统，并具备用于结构相似性评估的潜力，从而指导适应性调整。

3.2 一个示例：相似性评估

为便于说明，图1中给出了主要流程。左下角的方框展示了存储在案例库中的一个典型案例的图形化与属性化表示®。通过调用函数cover和copy，并结合适当的括号与逗号，我们能够以结构化的形式表达该案例的解：cover(copy(Y,3,Circle))。该表达式意指取一个圆，将其复制三次，并沿y轴方向排列；随后用4个单椭圆覆盖所有圆。右下角的方框则展示了待求解新问题的图形化与属性化表示。在同一方框中还给出了该问题的解，但目前尚不可用is。用户特别关注的目标（亦称子目标）是气源的连接。

对新问题的初步分析可通过一种基于属性描述的低成本表层相似性评估来完成。其结果是一组候选案例，这些案例在诸如方面、对象数量等表层属性上具有相似性。由此，存储于案例库中的庞大案例集可被缩减至少数几个有用的候选案例。

与例如机械工程不同，在机械工程中，功能依赖关系起着至关重要的作用，并且在推理过程中必须予以考虑。此外，还可以利用几何学中的相关规律（如对一个图形进行两次反射后将恢复为原图）。为简化起见，我们仅给出了每个对象的属性x、dx及其形态的取值。

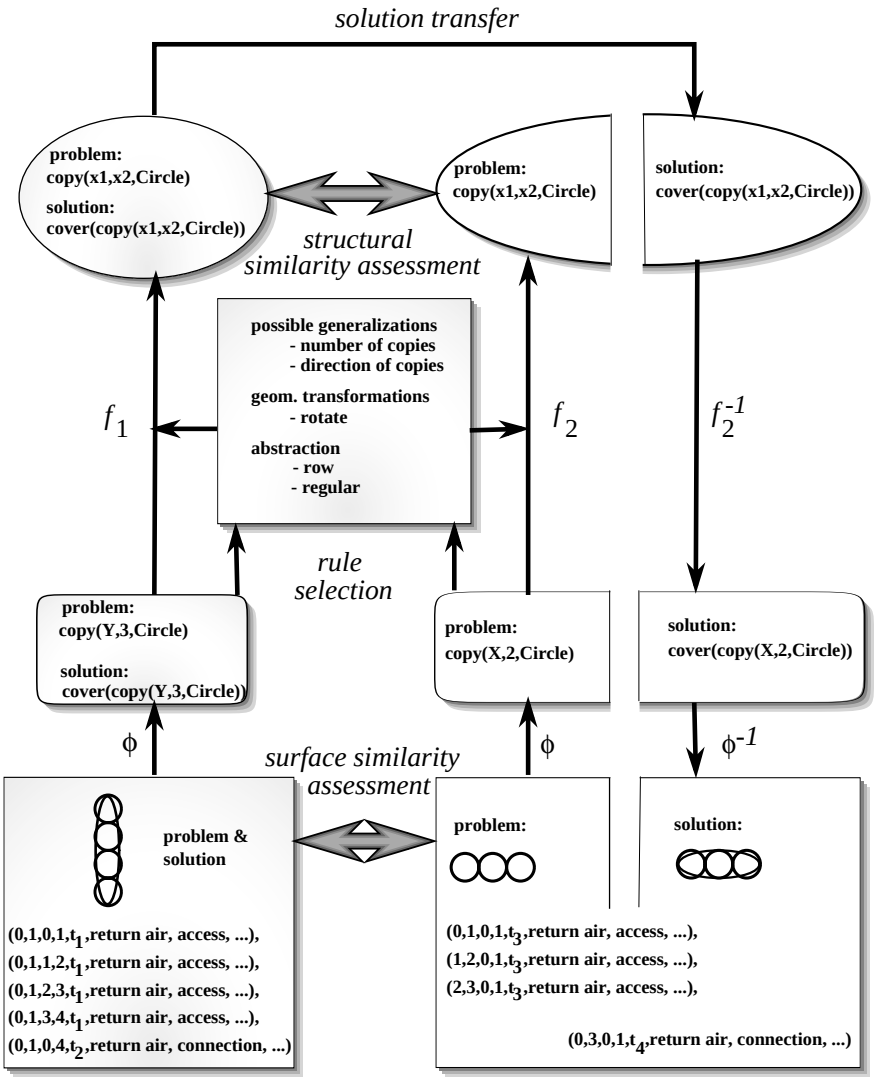


Fig. 3. An Example: Similarity Assessment and Adaptation

The next step is the transformation of the new problem into a structural representation. Here, candidate cases provide information about proper transformations referred to as ϕ. Thus, the new problem, which consists of three circles arranged in a row in the x-direction will be structurally represented by *copy(X, 2, Circle)*.

Based on the structural representation of superficially similar prior cases, the more expensive *structural similarity assessment* is performed. Axioms and

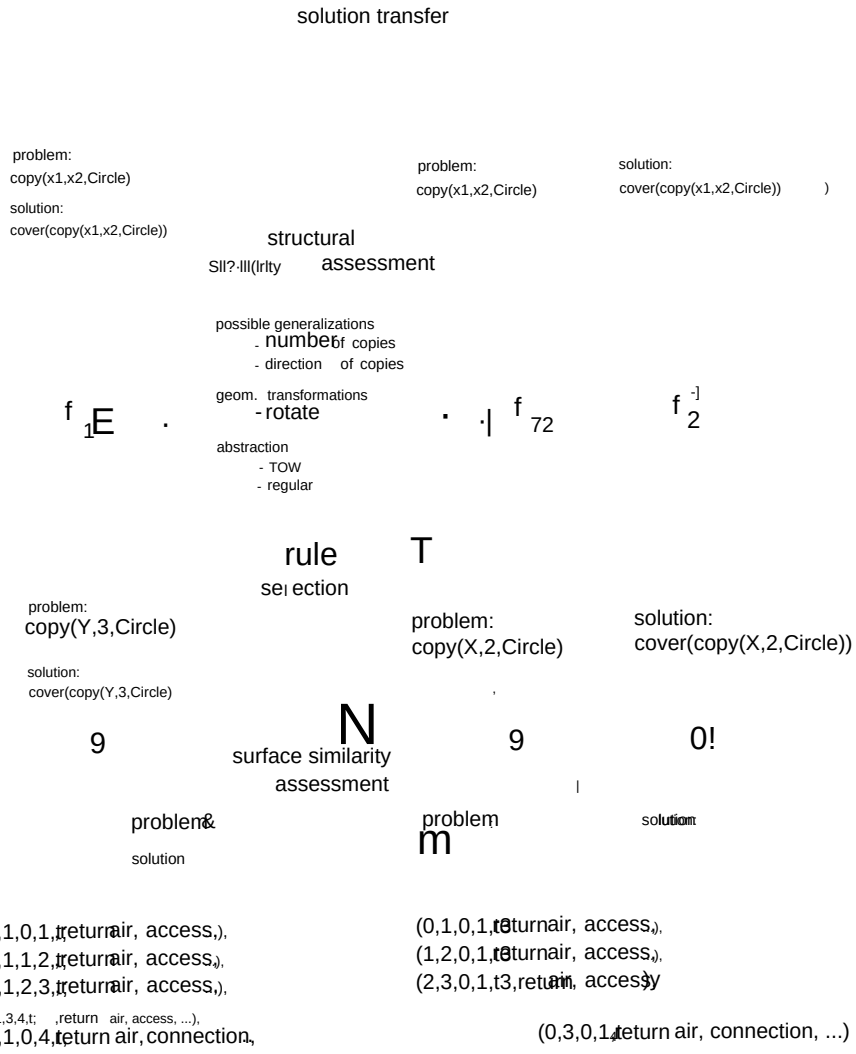


图3. 示例：相似性评估与适应

下一步是将新问题转化为结构化表示。在此，候选案例提供了关于适当变换的信息，这些变换记为¢。因此，由三个沿x方向排列成一排的圆构成的新问题，将被结构化地表示为copy(X,2,Circle))。

基于表面相似的先前案例的结构表示，进行更为昂贵的结构相似性评估。公理与

modification rules will be applied to determine the main structure which the new and a prior problem have in common. In our example, there are at least three different ways to achieve this:

- The first way uses generalization. For example, the concrete arrangement direction and the number of copies will be replaced by variables x_1 and x_2. The resulting common problem description will be *copy(x_1, x_2, Circle)* as shown in Fig. 3.
- The second way is to use generalization and geometrical transformation. Here, the number of copies will be generalized, too. One term representation will be rotated about 90 degrees.
- The third way uses abstraction. Here descriptions like *row* and *regular* will be used as abstract attributes. The idea behind this is that the more identical abstract attributes structural descriptions share, the more similar they are.

Given the main structure of both problem descriptions, we can simply transfer the also modified prior solution *cover(copy(x_1, x_2, Circle))* to the actual problem (in Fig. 3 referred to as *solution transfer*).

3.3 An Example: Adaptation

Using our knowledge about the sequence of modifications to determine the common structure, the transferred solution can be adapted to the new problem. This is denoted by f^{-1}. To get the concrete structural solution, in the example,

- where two generalizations were used to determine structural similarity, one has to replace x_1 by its former value X and analogously x_2 by 2. The resulting term will be *cover(copy(X,2,Circle))*.
- where generalization and geometrical transformation were used, one rotates the figure about -90 degrees (or 270 degrees) and replaces the variable number of copies by *2*. The resulting term will be *cover(copy(X,2,Circle))*.
- where abstraction was used, the transferred solution can be expressed by the attribute *covered*. The reverse concretization is somewhat difficult. Given terms and their corresponding abstract descriptions, one can try to find one structural representation which fulfils all abstract attributes (in this example *row, regular*, and *covered*). This suffices, if the number of these term-attribute assignments remains small but becomes intractable otherwise.

Given the structural representation of the new solution the application of the inverse transformation ϕ^{-1} yields the attribute-based and hence pictorial representation of the new solution.

4 Implementation

Our approach is prototypically implemented in SynTerm (for **Syn**thesis by using **Term**-based knowledge representations) and **SynGraph** (for **Syn**thesis by using

将应用修改规则以确定新问题与先前问题共有的主要结构。在我们的示例中，至少存在三种不同的实现方式：

e该first方法采用泛化处理。例如，具体的排列方向和副本数量将被变量21和5所替代。由此得到的通用问题描述如图3所示：copy(5l, 52, Circle)。

e方法second和ls to use generalization，以及副本数量的表示方式也将被推广。其中一个术语表示将旋转90度。e第三种方法采用抽象化处理。在此，诸如“行”和“规则”之类的描述将被用作抽象属性。其基本思想是：结构化描述所共享的抽象属性越相似，它们之间的相似性就越高。

鉴于两种问题描述的主体结构，我们可以将经过修改的先前解决方案覆盖（copy(ry1, x a, Circle)）直接迁移至实际问题中（如图3所示，称为解决方案迁移）。

3.3 一个示例：适应

利用我们对修饰顺序的了解来确定公共structure, the transferred solution，这一方法可以推广应用于新问题。这记作·

1. To get the concrete structural solution, in the example,。

在两种情况下，分别确定：其一，将21替换为其先前的值X；其二，类似地，将2替换为2。由此得到的项为cover(copy(X,2,Circle))。在另一处，通过使用泛化和几何变换，将图形绕原点旋转−90度（或270度），并将复制次数这一变量替换为2。由此得到的项仍为cover(copy(X,2,Circle))。在采用抽象化的情形下，所迁移的解决方案可通过属性“covered”来表达。而逆向的具体化则较为困难。给定若干项及其对应的抽象描述，可以尝试寻找一种结构化表示，该表示能够映射这些抽象属性（在本例中为row、regular和covered）。当此类项—属性对应关系的数量较少时，此方法尚可适用；但若数量较多，则会变得难以处理。

鉴于新解的结构化表示，应用逆变换b ¹即可得到该新解的基于属性的、进而为图示化的表示。

4 实施

我们的方法以原型形式实现了SynTerm（基于术语知识表示的合成）和SynGraph（基于图的合成）。

Graph-based representations). Both are integrated in a knowledge based system which operates in the domain of industrial building design. Whereas the overall system provides user interfaces and a broader range of functionality, SynTerm and SynGraph supply the system with solutions for special design problems. Thus SynTerm and SynGraph exchange solutions for problems of the system using the approach to similarity assessment and adaptation introduced. Fig. 4 illustrates the architecture of SynTerm and SynGraph. In the diagram, oval shapes represent data structures. Rectangular shapes represent processes.

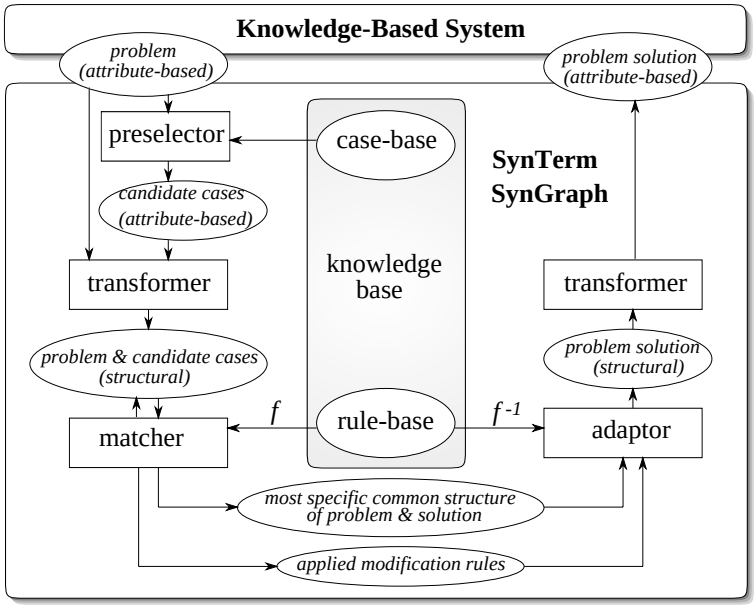


Fig. 4. Architecture of SynTerm and SynGraph

The CBR modules consist of a *preselector*, *transformers* (to transform the attribute-based representation into a structural one and vice versa), a *matcher*, and an *adaptor*. The knowledge base provides a canonical system of cases and rules. The input of SynTerm and SynGraph is an attribute-based problem description. Output is the corresponding solution in attribute-based representation. The first task is performed by the *preselector*. It defines a set of candidate cases from the case-base together with a suitable transformation function needed by the *transformer* to produce the structural problem description. The *matcher* applies rules to the current problem and the candidate cases until a most specific common structure of both can be determined. This most specific common structure of problems and the solution together with the applied modification rules are the input for the *adaptor*. In addition, the *adaptor* has access to the inverse

基于图的表示）。两者均集成于a知识库系统中，该系统应用于工业建筑设计领域。尽管整个系统提供了用户界面及更广泛的功能，但SynTerm和SynGraph则为系统提供针对特定设计问题的解决方案。因此，SynTerm和SynGraph采用所提出的相似性评估与适配方法，在系统内部交换问题求解方案。图4展示了SynTerm和SynGraph的架构；在该图中，椭圆形表示数据结构，矩形表示处理过程。

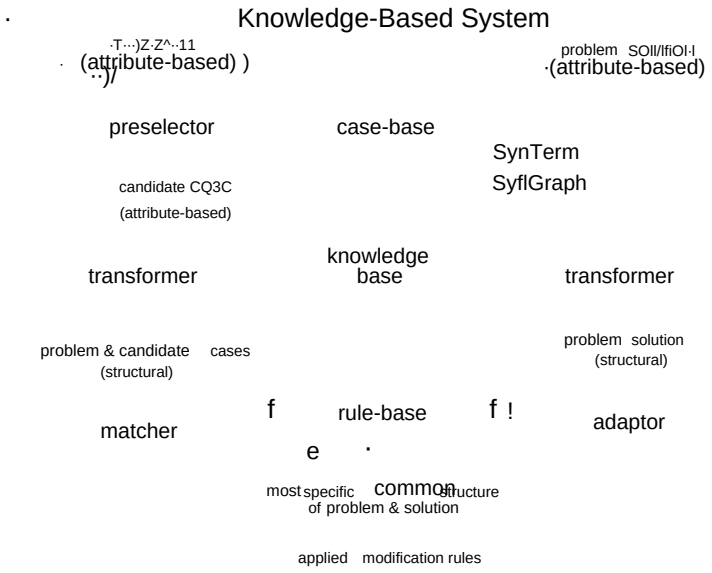


图4. SynTerm与SynGraph的架构

CBR模块由预选择器、变压器（用于在基于属性的表示与结构化表示之间进行转换）以及匹配器和适配器组成。知识库提供一套规范化的案例与规则体系。SynTerm和SynGraph的输入为基于属性的问题描述，输出则为相应的基于属性的解决方案。其中，预选择器负责执行第一项任务：它从案例库中确定一组候选案例，并同时指定变压器所需的应用变换函数，以生成结构化的问题描述。匹配器将规则应用于当前问题及各候选案例，直至能够确定两者间最具体的共同结构。该最具体的共同结构连同所应用的修改规则，作为适配器的输入；此外，适配器还可访问其逆变换函数。

modification rules stored in the rule-base. Output of the adaptor is the full specified problem solution in structural representation. The *transformer* provides its attribute-based representation.

Learning, e.g., the storage of new cases, new rules and the application of techniques like Knuth–Bendix completion to guarantee completeness of structural knowledge or learning of modification rule conditions or weights are not addressed in this implementation. This is one direction for future work.

The initial experimental results support the claims that the proposed approach offers improved adaptation facilities in synthesis tasks together with a greater overall explanation performance.

5 Conclusions

We presented a novel approach to similarity assessment and adaptation in CBR. Structural similarity is assessed if and only if there is knowledge available to adapt old solutions correctly to the current problem of interest. Background knowledge is organized in a manner that permits the efficient identification of the appropriate cases and modification rules. The result is an approach to structural similarity which implies adaptability. Retrieved cases are adaptable cases. Only one similar, adaptable case is needed and looked for.

The integration of retrieval, matching and adaptation essentially improves case-based problem solving, especially for synthesis tasks where adaptation is important. In addition, the structural similarity assessment provides a basis for more descriptive explanations for why particular solutions have been selected and adapted.

The basic mechanisms of our approach are domain independent and thus facilitate the adoption of the technique across a range of CBR application domains like software design, machinery, and technical configuration. Detailed investigations are under work.

6 Acknowledgements

This research has been strongly inspired by work done in the project FABEL the general objective of which is the integration of case-based and model-based approaches in knowledge-based systems. This work was guided by BIPIN INDURKHYA's work on analogical reasoning. He contributed to this work by providing detailed and valuable comments. I wish to thank DIETMAR JANETZKO for his helpful comments on an earlier draft. I am indebted to my thesis advisor KLAUS P. JANTKE for his continuing support and encouraging feedback. PHIL WATSON deserves thanks for his aid in revising an earlier version of this paper. Nonetheless, the paper reflects our personal view.

修改 rules stored in the rule-base. 适配器的输出是以结构化表示形式给出的完整指定问题解。变换器则提供其基于属性的表示。

在本实现中，尚未涉及诸如新案例与新规则的存储、以及应用如Knuth-Bendix归约等技术以确保结构知识的完备性，或学习修改规则的条件与权重等内容。这将是未来研究的一个重要方向。

初步实验结果证实了所提出的方案在合成任务中能够提供更优的适应性，并显著提升整体的解释性能。

5 结论

我们提出了一种用于案例推理中相似性评估与适应的新方法。只有在具备能够将旧解决方案正确适配至当前待解决问题的知识时，才会进行结构相似性的评估。背景知识以一种有序的方式组织，从而能够高效地识别出适当的案例并实施相应的修改。由此形成了一种将适应性内置于其中的结构相似性评估方法。被检索出的案例即为可适应的案例。在该方法中，只需寻找并获取一个相似且可适应的案例。

检索、匹配与适应的集成在本质上提升了基于案例的问题求解能力，尤其是在需要进行适应的综合任务中，适应环节尤为重要。此外，结构相似性评估为解释为何选择并调整特定解决方案提供了更为详尽的依据。

我们方法的基本机制具有领域无关性，因此有助于在软件设计、机械工程以及技术配置等一系列基于案例推理的应用领域中推广应用该技术。相关深入研究仍在进行中。

6 致谢

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